

containing page, will not have their data retrieved.

Before the arrival of the Canvas API, it was not possible to programmatically retrieve the pixel values of a downloaded image. Private images from other sites could be displayed in a page but not read or copied. Allowing scripts to read image data from other origins would effectively share users' photographs and other sensitive online image file with the entire web.

In order to prevent this, any canvas that contains images rendered from remote origins will throw a security exception if the `getImageData` function is called. It is perfectly acceptable to render remote images into a canvas from another origin as long as you do not attempt to fetch the data from that canvas after it has been tainted.

SVG

Scalable Vector Graphics (SVG) integrates with the HTML5 and CSS features to unleash some of the most beautiful experiences on the web.

The format makes use of specific XML tags to generate a vector-based image. This makes SVG more easily read and edited by hand than other graphic formats. And because an SVG file is composed entirely of markup, text within the vector image is searchable. Moreover, SVG benefits from the richness of having a DOM. Images are scriptable graphic objects, enabling much more powerful web applications. It's possible to create dynamically generated images without requiring server-side scripting or browser plug-ins.

SVG offers many distinct advantages over other formats. SVG images scale up with little increase in file size and with no loss of fidelity, making it ideal for use in smaller images and icons. SVG is also a fantastic format for images with larger dimensions. Maps, charts, and diagrams are examples. Small labels and detailed borders remain intact even when images are scaled down. Key areas of focus, like the text in an organization chart or the border of a country, remain crisp and readable after zooming in on the image. The preservation of information in human-readable markup also makes SVG suitable for Wiki-type collaborative images, as it is easily edited. SVG makes changing labels in charts and diagrams a breeze – only a basic text editor is needed. As native SVG support in web browsers continues to broaden, SVG becomes an increasingly good option for implementing interactive and animated web content.

In HTML5, SVG can be included as inline HTML. Current webpages can be updated to include inline SVG with little alteration to their overall